33. Review 2024 Exam

April 8, 2026 9:25 AM

1. Short Answer Questions (20 pt)

(i) Briefly explain the difference between bootstrapping and cross validation.

(ii) Briefly explain the difference between the Robbins-Monro and the Newton-Raphson algorithms. To help, the formulae for updating the parameters are

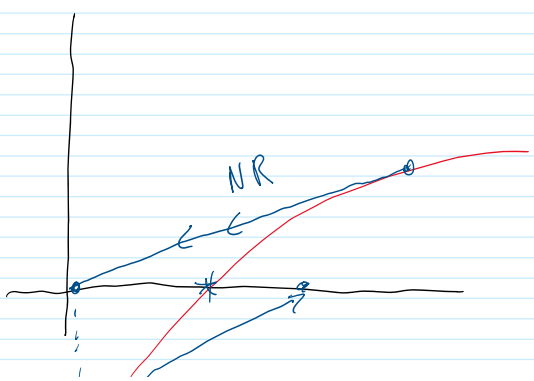
$$\text{RM: } \theta_{i+1} \leftarrow \theta_i - a_k y(\theta_i)$$
$$\text{NR: } \theta_{i+1} \leftarrow \theta_i - f(\theta_i)/f'(\theta_i)$$

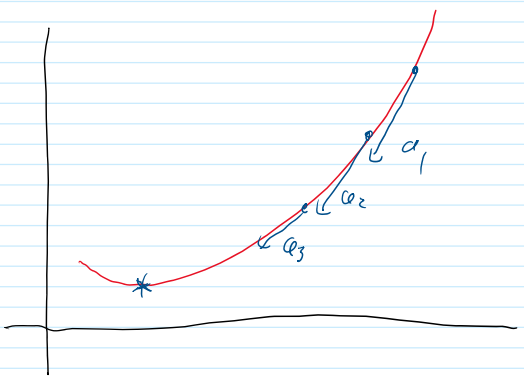
(i) Both are methods for sampling from a data set but with different goals in mind.

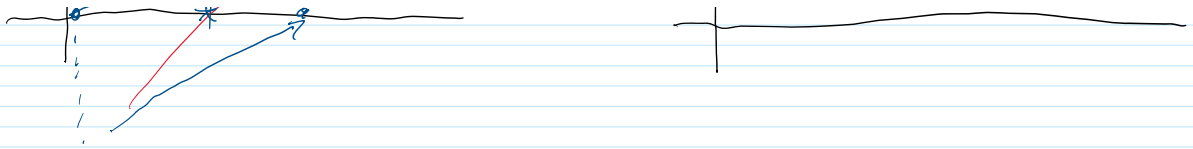
- bootstrapping is when we resample from our data to get new datasets which are often used to estimate uncertainty, e.g. confidence interval for an unknown parameter.
- CV is when we split our data into training + testing sets with the goal of choosing optimal values for tuning parameters.

(ii) Both are in a sense optimizing a function. However, RM tries to find a minimum where NR looks for a root, $f(x)=0$

In RM, we have to choose a step size where as in NR, we follow the tangent line to the horizontal axis.







2. Transformation of Random Variables (20 pt)

Let $X, Y \sim$ Exponential (1). Find the distribution (or density) function of

iid

$$Z = \frac{X}{X+Y}.$$

Hint: Let $W = X + Y$.

Change $\begin{pmatrix} x \\ y \end{pmatrix} \to \begin{pmatrix} z \\ w \end{pmatrix}$ Step 1: write X, Y in terms of z, w

$$z = \frac{x}{x+y} = \frac{x}{w} \Rightarrow x = zw$$

$$w = x + y, \quad y = w - x \Rightarrow y = w - zw = w(1-z)$$

Step 2: Jacobian: $\frac{\partial x}{\partial z}, \frac{\partial x}{\partial w}, \frac{\partial y}{\partial z}, \frac{\partial y}{\partial w}$

$$|J| = \begin{vmatrix} w & z \\ -w & 1-z \end{vmatrix} = w(1-z) + wz$$

$$w - wz + wz = \boxed{w}$$

Step 3: Change the variables!

$$f_{zw}(z, w) = f_{x,y}(x(z,w), y(z,w))\,|J|$$

$f_{xy}(x,y) = e^{-x} e^{-y}$ ← product of 2 exponential pdfs (Independence)

$$= \exp(-zw)\exp(-w(1-z))\,|w|$$

$w \geq 0$

$$= |w| \exp(-zw - w + zw)$$

FYI w is Gamma

$$= w e^{-w} \times 1$$

$z = \frac{x}{x+y} \in [0, 1]$ Uniform distribution!

we found that $f_{zw}(z,w)$ is constant in z

Here, we get that z and w are also indep r.v.s However, sometimes you have to integrate out w to find the density of z

out w to find the density of z

$$f_z(z) = \int f_{zw}(z, w)\, dw$$

3. Uniform or Importance Sampling (20 pt)

We want to estimate

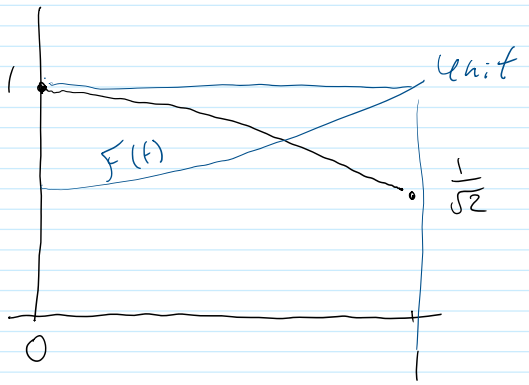
$$\int_0^1 g(t)dt \quad \text{with} \quad g(t) = \frac{1}{\sqrt{1+t^2}}.$$

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Uniform}\,[0,1]$ and $Y_1, \ldots, Y_n \overset{\text{iid}}{\sim} f(t)$ where

$$f(t) = \frac{1}{\log(2)} \frac{1}{2-t}$$

on $[0,1]$. Determine which of $n^{-1}\sum_{i=1}^n g(X_i)$ and $n^{-1}\sum_{i=1}^n g(Y_i)/f(Y_i)$ has the smaller variance.[1]

$\boxed{\text{Uniform}}$



$$\hat{\mu} = \frac{1}{n} \sum g(X_i)$$

$$Var(\hat{\mu}) = \frac{1}{n} Var(g(X))$$

$$\int_0^1 g(t)^2 dt = \int_0^1 \frac{1}{1+t^2} dt$$

$\boxed{\text{Importance}}$

$$\int_0^1 \frac{g(t)^2}{f(t)} \cancel{f(t)}\, dt = \int_0^1 \frac{1}{1+t^2} \cdot (2-t)\, dt \quad (\text{by } 2)$$